

CS & IT ENGINEERING



Computer Network

Error Control

Lecture No. - 08



By - Abhishek Sir



Recap of Previous Lecture



Topic

Hamming Code

Topic

Hamming Distance



Topics to be Covered



Topic

Hamming Distance

Topic

Bit Error Probability

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW



[MCQ]

[GATE-2021][2 Mark]



#Q. Assume that a 12-bit Hamming codeword consisting of 8-bit data and 4 check bits is $d_8d_7d_6d_5c_8d_4d_3d_2c_4d_1c_2c_1$, where the data bits and the check bits are given in the following tables:

Data bits							
d_8	d_7	d_6	d_5	d_4	d_3	d_2	d_1
1	1	0	x	0	1	0	1

Check bits			
c_8	c_4	c_2	c_1
y	0	1	0

Which of the following choices gives the correct values of x and y?

A

x is 0 and y is 0

B

x is 0 and y is 1

C

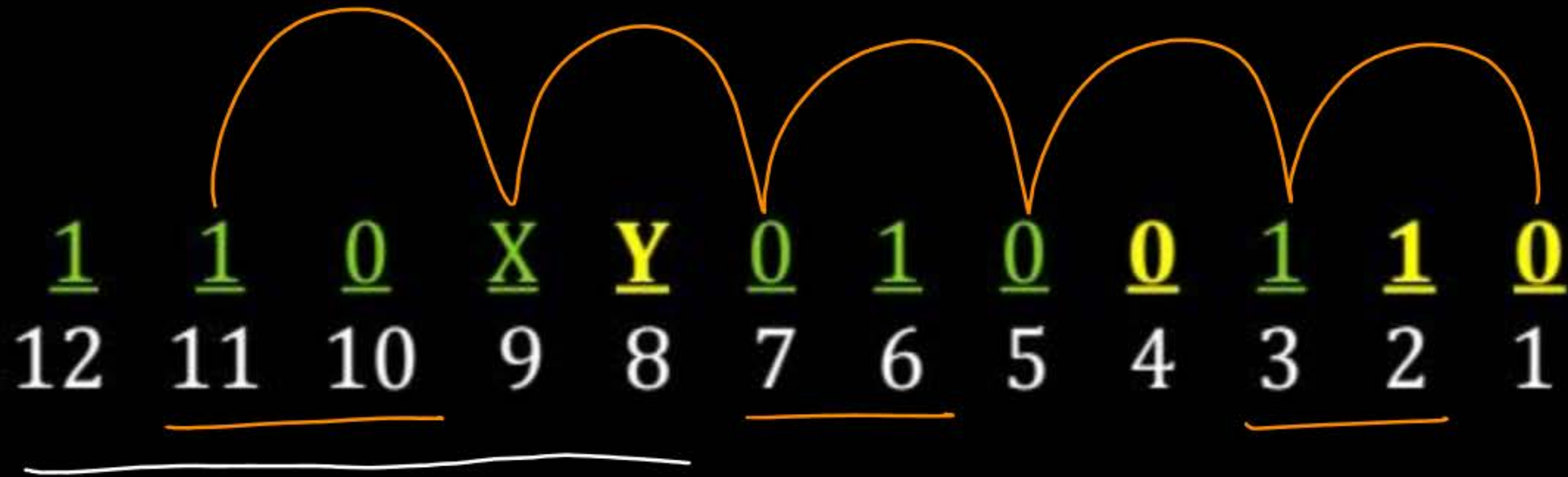
x is 1 and y is 0

D

x is 1 and y is 1

Ans: A

Even or odd parity = ? $\Rightarrow$ Even Parity



X should be zero.

Y should be zero

Solution :-

<u>1</u>	<u>1</u>	<u>0</u>	<u>X</u>	<u>Y</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>
12	11	10	9	8	7	6	5	4	3	2	1

=> Identify even or odd parity

→ Parity R_1 is balanced using “even parity”

=> Identify value of variable X

→ As per parity R_0 , X should be “zero bit”

=> Identify value of variable Y

→ If X is “zero”, then as per parity R_3 , Y should be “zero bit”



Topic : Hamming Weight



→ The weight of a codeword is the number of nonzero elements

Hamming weight of a binary string = Number of one's in that string

Example :-

Hamming Weight["1011101"] = 5



Topic : Hamming Distance



→ Metric for comparing two binary strings of equal length

Hamming distance between two binary strings of equal length

= Number of positions at which the corresponding bits are different



Topic : Hamming Distance



→ Let suppose A and B are two binary strings of equal length

$$\begin{aligned} \underbrace{d(A,B)} &= \text{Hamming Distance between } A \text{ and } B \\ &= \text{Hamming Weight of } [A \text{ bit-wise XOR } B] \end{aligned}$$



Topic : Hamming Distance



Example :- A = 1011101 and B = 0110110

bit-wise XOR A = 1 0 1 1 1 0 1
 B = 0 1 1 0 1 1 0

Result = 1 1 0 1 0 1 1

$d(A,B) = \text{Hamming Weight}[\text{Result}] = 5$



Topic : Hamming Distance



Let suppose set of valid codewords :

Codeword C_1 : 00000 00000

Codeword C_2 : 00000 11111

Codeword C_3 : 11111 00000

Codeword C_4 : 11111 11111

$\min^m \text{H.D.} = 5$

minimum Hamming Distance =

= Minimum [$d(C_1, C_2)$, $d(C_1, C_3)$, $d(C_1, C_4)$, $d(C_2, C_3)$, $d(C_2, C_4)$, $d(C_3, C_4)$]

= Min [5, 5, 10, 10, 5, 5]

= 5



Topic : Hamming Distance



CASE I : No any error ✓

Transmitted Code = 00000 00000 (Valid Codeword) ✓

Received Code = 00000 00000 (Valid Codeword)

Error Detected : "No" any Error detected
Corrected Codeword :

Codeword C_1 : 00000 00000

Codeword C_2 : 00000 11111

Codeword C_3 : 11111 00000

Codeword C_4 : 11111 11111



Topic : Hamming Distance



CASE II : One-bit error ✓

⇒ Detect and correct successfully.

Transmitted Code = 00000 00000 (Valid Codeword) ✓

Received Code = 00000 00001 (Invalid Codeword)

Error Detected : "Yes"

Corrected Codeword : 00000 00000 (Nearest Valid Codeword)

Codeword C_1 : 00000 00000

Codeword C_2 : 00000 11111

Codeword C_3 : 11111 00000

Codeword C_4 : 11111 11111

$$d(C_1, R_C) = 1$$

$$d(C_2, R_C) = 4$$

$$d(C_3, R_C) = 6$$

$$d(C_4, R_C) = 9$$



Topic : Hamming Distance



CASE III : Two-bit error

→ Detected and corrected successfully

Transmitted Code = 00000 00000 (Valid Codeword) ✓

Received Code = 00000 00011 (Invalid Codeword)

Error Detected : "Yes"

Corrected Codeword : 00000 00000 (Nearest Valid Codeword)

Codeword C_1 : 00000 00000	$d(C_1, R_C) = 2$
Codeword C_2 : 00000 11111	$d(C_2, R_C) = 3$
Codeword C_3 : 11111 00000	$d(C_3, R_C) = 7$
Codeword C_4 : 11111 11111	$d(C_4, R_C) = 8$



Topic : Hamming Distance



CASE IV : Three-bit error

→ Detected successfully
but not corrected successfully.

Transmitted Code = 00000 00000 (Valid Codeword)

Received Code = 00000 00111 (Invalid Codeword)

Error Detected : "Yes"

Corrected Codeword : 00000 11111 (Nearest Valid Codeword)

Codeword C_1 :	00000 00000	$d(C_1, RC) = 3$
Codeword C_2 :	00000 11111	$d(C_2, RC) = 2$
Codeword C_3 :	11111 00000	$d(C_3, RC) = 8$
Codeword C_4 :	11111 11111	$d(C_4, RC) = 7$



Topic : Hamming Distance



CASE V : Four-bit error

→ Detected successfully
but not corrected successfully.

Transmitted Code = 00000 00000 (Valid Codeword)

Received Code = 00000 01111 (Invalid Codeword)

Error Detected : "Yes"

Corrected Codeword : 00000 11111 (Nearest Valid Codeword)

Codeword C_1 : 00000 00000	$d(C_1, RC) = 4$
Codeword C_2 : 00000 11111	$d(C_2, RC) = 1$
Codeword C_3 : 11111 00000	$d(C_3, RC) = 9$
Codeword C_4 : 11111 11111	$d(C_4, RC) = 6$



Topic : Hamming Distance



Not detecting Error

CASE VI : **Five-bit error**

Transmitted Code = **00000 00000** (Valid Codeword)

Received Code = **00000 11111** (Valid Codeword)

Error Detected : **"No"** any Error detected
Corrected Codeword :

Codeword C_1 : 00000 00000

Codeword C_2 : 00000 11111

Codeword C_3 : 11111 00000

Codeword C_4 : 11111 11111



Topic : Hamming Distance



CASE VII : Five-bit error

Transmitted Code = 00000 00000 (Valid Codeword)

Received Code = 00001 01111 (Invalid codeword)

Error Detected : "Yes"

Corrected Codeword : 00000 11111 (Nearest Valid Codeword)

Codeword C_1 : 00000 00000

Codeword C_2 : 00000 11111

Codeword C_3 : 11111 00000

Codeword C_4 : 11111 11111



Topic : Hamming Distance

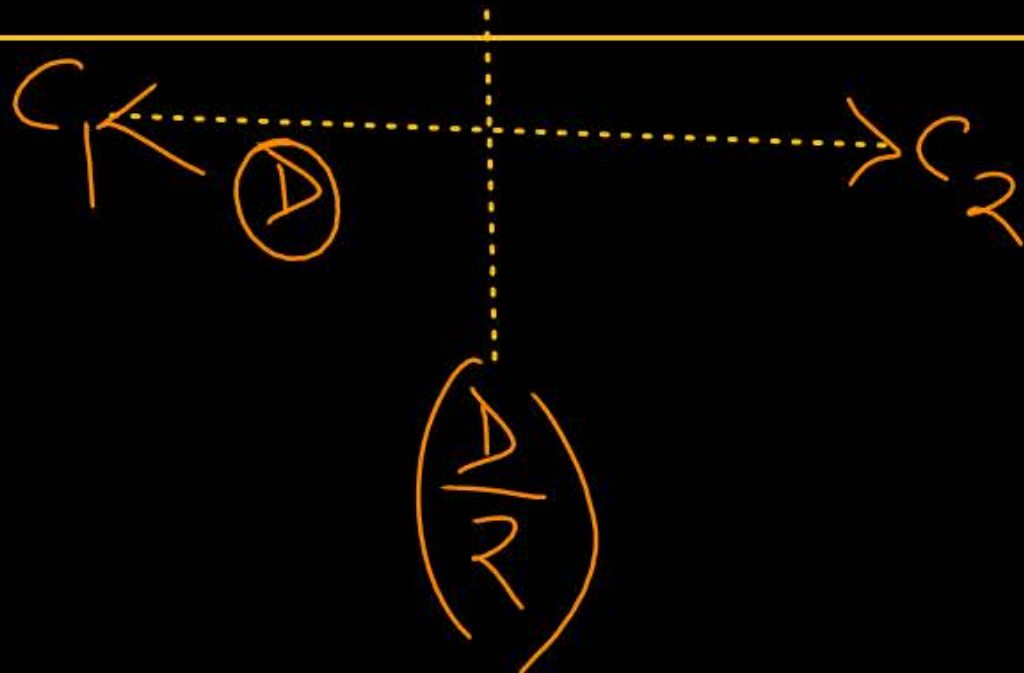


if (minimum) Hamming Distance is D

then receiver can detect upto $(D-1)$ bits error

and receiver can correct upto $\text{Floor}[(D-1)/2]$ bits error

$$\left\lfloor \frac{(D-1)}{2} \right\rfloor$$





Topic : Hamming Distance ✓



To detect up to x - bits error

-> minimum Hamming Distance should be $(x+1)$

To correct up to y - bits error

-> minimum Hamming Distance should be $(2y+1)$

#Q. An error correcting code has the following code words:

c_1 c_2 c_3 c_4 c_5
 00000000, 00001111, 01010101, 10101010, 11110000

What is the maximum number of bit errors that can be corrected?

$$\min^m H.D.(D) =$$

$$\min [d(c_1, c_2), d(c_1, c_3), d(c_1, c_4), d(c_1, c_5), \\ d(c_2, c_3), d(c_2, c_4), d(c_2, c_5), d(c_3, c_4), \\ d(c_3, c_5), d(c_4, c_5)]$$

$$D = \min^m [4, 4, 4, 4, 4, 4, 8, 8, 4, 4]$$

$$D = 4$$

[GATE 2007]

IIT-K

(A) 0

✓ (B) 1

(C) 2

(D) 3

Ans: B

Solution :-

$$C_1 = 00000000$$

$$C_2 = 00001111$$

$$C_3 = 01010101$$

$$C_4 = 10101010$$

$$C_5 = 11110000$$

minimum Hamming Distance (D) =

$$\text{Minimum} [d(C_1, C_2), d(C_1, C_3), d(C_1, C_4), d(C_1, C_5), \\ d(C_2, C_3), d(C_2, C_4), d(C_2, C_5), \\ d(C_3, C_4), d(C_3, C_5), d(C_4, C_5)]$$

→ $D = 4$

→ Receiver can correct upto $\text{Floor}[(D-1)/2]$ bits error

$$\left\lfloor \frac{(D-1)}{2} \right\rfloor = \left\lfloor \frac{(4-1)}{2} \right\rfloor = \left\lfloor \frac{3}{2} \right\rfloor = 1$$

#Q. Consider a binary code that consists only four valid codewords as given below.

00000, 01011, 10101, 11110

c_1 c_2 c_3 c_4

Let minimum Hamming distance of code be p and maximum number of erroneous bits that can be corrected by the code be q . The value of p and q are:

[GATE 2017]

IIT-R

$\min^m H.D. (P)$

$= \min [d(c_1, c_2), d(c_1, c_3),$

$d(c_1, c_4), d(c_2, c_3),$

$d(c_2, c_4), d(c_3, c_4)]$

$P = \min [3, 3, 4, 4, 3, 3]$

$P = 3$

☒ (A) $p = 3$ and $q = 1$

☐ (B) $p = 3$ and $q = 2$

☐ (C) $p = 4$ and $q = 1$

☐ (D) $p = 4$ and $q = 2$

Ans: A

Solution :-

$$C_1 = 00000$$

$$C_2 = 01011$$

$$C_3 = 10101$$

$$C_4 = 11110$$

minimum Hamming Distance (p) =

$$\text{Minimum} [d(C_1, C_2), d(C_1, C_3), d(C_1, C_4), \\ d(C_2, C_3), d(C_2, C_4), d(C_3, C_4)]$$

$$\rightarrow p = 3$$

Receiver can correct upto $\text{Floor}[(p-1)/2]$ bits error

$$\rightarrow q = \text{Floor}[(p-1)/2]$$

$$\rightarrow q = 1$$

$$\left\lfloor \frac{(p-1)}{2} \right\rfloor = \left\lfloor \frac{(3-1)}{2} \right\rfloor = 1$$



Topic : Cyclic Code



One-bit parity (with even parity) and 3 data bits

Data --> Codeword
 $d_2d_1d_0$ --> $d_2d_1d_0P$

$$\min^m H.D. = 2$$

0 0 0 --> 0 0 0 0
0 0 1 --> 0 0 1 1
0 1 0 --> 0 1 0 1
0 1 1 --> 0 1 1 0
1 0 0 --> 1 0 0 1
1 0 1 --> 1 0 1 0
1 1 0 --> 1 1 0 0
1 1 1 --> 1 1 1 1

$\Rightarrow$ Non-zero codeword
 $\min^m \text{weight} = 2$

2^3



Topic : Linear Code



Hamming Code (with Even Parity) and 3 data bits :

Data → Codeword
 $d_2d_1d_0$ → $d_2d_1C_2d_0C_1C_0$

$\min^m H.D. = 3$

0 0 0	→	0 0 0 0 0 0
0 0 1	→	0 0 0 1 1 1
0 1 0	→	0 1 1 0 0 1
0 1 1	→	0 1 1 1 1 0
1 0 0	→	1 0 1 0 1 0
1 0 1	→	1 0 1 1 0 1
1 1 0	→	1 1 0 0 1 1
1 1 1	→	1 1 0 1 0 0

Non-zero codeword
 $\min^m \text{weight} = 3$

2^3



Topic : Linear Code

$$\text{Min}^m \text{H.D} = 4$$



Two-dimension Parity (with Even Parity) and 4 data bits :

Data → Codeword
 $d_3 d_2 d_1 d_0 \rightarrow d_3 d_2 \underline{R_1} d_1 d_0 \underline{R_2} \underline{C_1} \underline{C_2} \underline{P}$

0 0 0 0	→	0 0 0 0 0 0 0 0
0 0 0 1	→	0 0 0 1 1 0 1 1
0 0 1 0	→	0 0 1 0 1 1 0 1
0 0 1 1	→	0 0 1 1 0 1 1 0
0 1 0 0	→	0 1 1 0 0 0 1 1
0 1 0 1	→	0 1 1 0 1 1 0 0
0 1 1 0	→	0 1 1 1 0 1 1 0
0 1 1 1	→	0 1 1 1 1 0 1 0

Data → Codeword
 $d_3 d_2 d_1 d_0 \rightarrow d_3 d_2 \underline{R_1} d_1 d_0 \underline{R_2} \underline{C_1} \underline{C_2} \underline{P}$

1 0 0 0	→	1 0 1 0 0 0 1 0
1 0 0 1	→	1 0 1 0 1 1 1 0
1 0 1 0	→	1 0 1 1 0 1 0 0
1 0 1 1	→	1 0 1 1 1 0 0 1
1 1 0 0	→	1 1 0 0 0 0 1 1
1 1 0 1	→	1 1 0 0 1 1 1 0
1 1 1 0	→	1 1 0 1 0 1 0 1
1 1 1 1	→	1 1 0 1 1 0 0 0



Topic : Hamming Distance



(minimum) Hamming Distance of the linear code is
= minimum weight of its nonzero codewords



Topic : Hamming Distance

→ (minimum) Hamming Distance in:

1. One-bit parity is always **"2"**.
→ Always detect "single bit error"

2. Hamming Code is always **"3"**.
→ Always correct "single bit error"
→ Always detect "upto two bit error"

3. Two-dimension parity is always **"4"**.
→ Always correct "single bit error"
→ Always detect "upto three bit error"



Topic : Block Code



Set of Codewords in :

- CRC is “Cyclic Code”
- Hamming Code is “Linear Code, but not Cyclic”
- Two-dimension parity is “Linear Code, but not Cyclic”



Topic : Bit Error Rate



=> Bit Error Rate (BER)

→ Number of bit errors per unit time

=> Bit Error Ratio or Bit Error Probability (P_b)

→ Probability that a bit is corrupted

Example 1:-

#Q. Consider source and destination hosts are directly connected through a point-to-point noisy communication link. A frame of length L bits is transmitted into the channel and the probability that a bit of the frame gets flipped during transmission (bit error probability) is P_b . What is the probability that the frame is delivered error-free to the destination host?

- A $(P_b)^L$
- ☒ B $(1 - P_b)^L$
- C $P_b * (1 - P_b)^{(L-1)}$
- D $1 - (1 - P_b)^L$

Bit Error Prob. = P_b , Frame length = L bits

$$\begin{aligned} \text{Ans} &= (1 - P_b) \times (1 - P_b) \times \dots \times (1 - P_b) \\ &= (1 - P_b)^L \end{aligned}$$

Ans: B

[NAT]

IIT-R, H.W.

[GATE-2025, Set-1] [2 Mark]



#Q. Suppose a 5-bit message is transmitted from a source to a destination through a noisy channel. The probability that a bit of the message gets flipped during transmission is 0.01. Flipping of each bit is independent of one another. The probability that the message is delivered error-free to the destination is _____. (rounded off to three decimal places)

#Q. In a communication network, a packet of length L bits takes link $L1$ with a probability of p_1 or link $L2$ with a probability of p_2 . Link $L1$ and $L2$ have bit error probability of b_1 and b_2 respectively. The probability that the packet will be received without error via either $L1$ or $L2$ is

[GATE 2005]

IIT-B
H.W.

A $(1 - b_1)^L p_1 + (1 - b_2)^L p_2$

B $[1 - (b_1 + b_2)^L] p_1 p_2$

C $(1 - b_1)^L (1 - b_2)^L p_1 p_2$

D $1 - (b_1^L p_1 + b_2^L p_2)$

Example 2:-



#Q. Consider source and destination hosts are directly connected through a point-to-point noisy communication link. A frame of length L bits is transmitted into the channel and the probability that a bit of the frame gets flipped during transmission (bit error probability) is P_b . What is the probability that the frame is delivered to the destination host contains some error(s)?

- A $(P_b)^L$
- B $(1 - P_b)^L$
- C $P_b * (1 - P_b)^{(L-1)}$
- ☒ D $1 - (1 - P_b)^L$

Prob. that frame delivered error-free
 $= (1 - P_b)^L$

$$\text{Ans} = 1 - (1 - P_b)^L$$

Ans: D



Topic : Packet Error Probability

=> Packet Error Probability (P_p)
→ Probability that a packet is corrupted
[Packet contains corrupted bits]

Suppose P_b = bit error probability

and Frame Length = L bits

$$\text{Packet error probability } (P_p) = 1 - (1 - P_b)^L$$



2 mins Summary



Topic

Hamming Distance ✓

Topic

Bit Error Probability ✓

Imp.

CRC

Hamming Dist.



THANK - YOU

